

ABC analysis — lecture example (10 part numbers)

Part 4 · How do we begin? — solution report with data, method, results and charts

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1. ABC analysis — lecture example (10 part numbers)

Step 1 — annual dollar usage

Annual usage analysis (slide 32).

part number	annual unit usage	unit cost	annual dollar usage
1	1,100	\$2	\$2,200
2	600	\$40	\$24,000
3	100	\$4	\$400
4	1,300	\$1	\$1,300
5	100	\$60	\$6,000
6	10	\$25	\$250
7	100	\$2	\$200
8	1,500	\$2	\$3,000
9	200	\$2	\$400
10	500	\$1	\$500

\$38,250

Total annual \$ usage

10

Number of parts

Step 2 — rank, cumulate and classify

Ranking by annual \$ usage (slide 34) with classes from both cut-off rules.

part number	annual dollar usage	cum value	cum pct value	cum pct items	class	class by value 80 95
2	\$24,000	\$24,000	62.7%	10%	A	A
5	\$6,000	\$30,000	78.4%	20%	A	A
8	\$3,000	\$33,000	86.3%	30%	B	B
1	\$2,200	\$35,200	92.0%	40%	B	B
4	\$1,300	\$36,500	95.4%	50%	B	C
10	\$500	\$37,000	96.7%	60%	C	C
9	\$400	\$37,400	97.8%	70%	C	C
3	\$400	\$37,800	98.8%	80%	C	C
6	\$250	\$38,050	99.5%	90%	C	C
7	\$200	\$38,250	100.0%	100%	C	C

With the **items rule (20 % / 30 % / 50 %)** parts 2 and 5 are A, parts 8, 1, 4 are B and the rest C — A holds **78 %** of the value with 20 % of the items, exactly the Pareto pattern on slide 30. With **strict value cut-offs (80 % / 95 %)** part 4 (cumulative 95.4 %) falls just into C: items at a boundary are a **judgement call** — managers often promote them for other reasons (criticality, scarcity).

Pareto chart — lecture example

Part 2 alone is 63 % of annual \$ usage

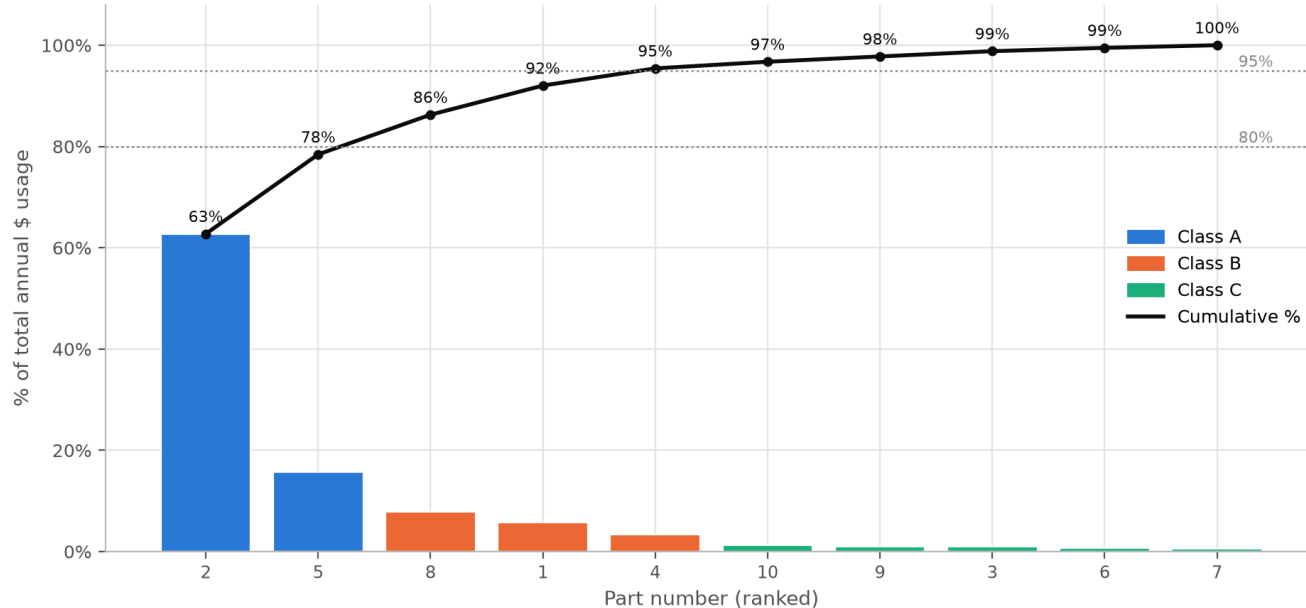


Figure 1.1 — Bars = % of annual \$ usage per part (coloured by class); line = cumulative %.

ABC curve

Cumulative % of items vs cumulative % of value

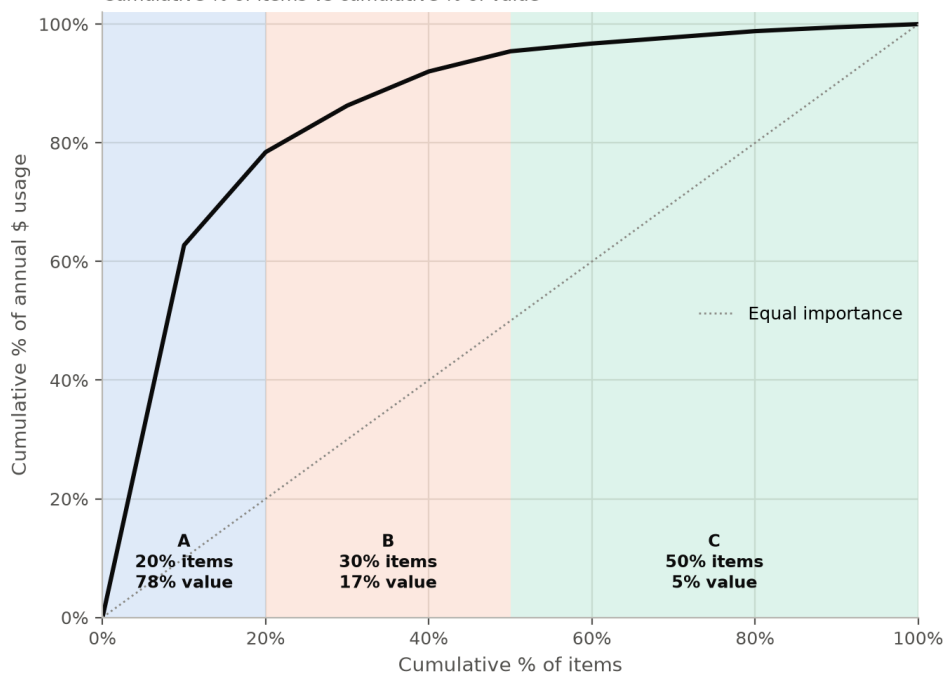


Figure 1.2 — ABC (Lorenz) curve with class bands.

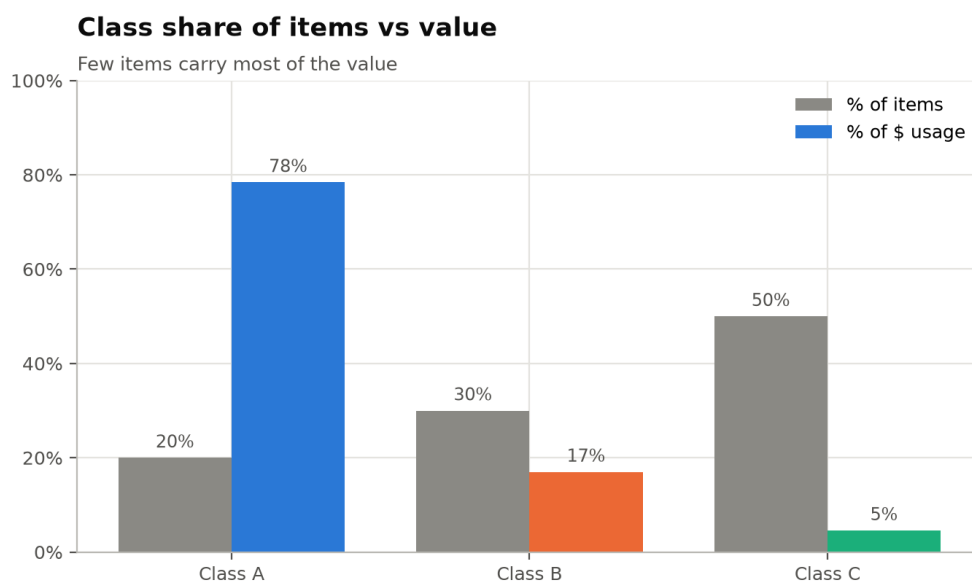


Figure 1.3 — Share of items and value by class.

Class summary (items rule).

class	items	value	pct items	pct value
A	2	\$30,000	20%	78.4%
B	3	\$6,500	30%	17.0%
C	5	\$1,750	50%	4.6%